

NEW · SPARMANIK FARM · STAGE-BY-STAGE PLAYBOOK

NEW SOP *Melon* Book

Grow it one stage at a time.
Read only the page you're on.

Target 1.8–2.2 kg · brix 15+



 Prep → Nursery → Grow → Harvest

 Set-&-forget feed & water

Sparmanik Farm · Indonesian Farm · Kab. Simalungun · v3.7 · 6 Aug 2026

HOW THE CROP RUNS

The stages

★	START HERE — make the feed	3	W	Watering — Day 0–25	14
1	Nursery — seed to seedling	4	4	Flower & set (HST 25–42)	19
2	Prep — buffer coir, bags, house	6	5	Fruit sizing (HST 42–60)	23
T	Transplant + Trichoderma drench	8	6	Ripen & finish (HST 60–80)	26
M	Make Stock A & B (night before)	11	7	Harvest & turnaround	29
3	Establish & grow (HST 0–25)	13	S	Crop schedule (year plan)	30
			R	Reference (recipe, stock, checks)	35

≡ How to use this book

Find today's stage. Read *only* that page — it has what to do, feed & water, spray, what to watch for, and when to move on. The **Reference** at the back (exact grams, stock-making, ppm, checks, pictures) is for whoever makes up the drums.

START HERE · WHAT MOST DAYS NEED

Make the feed

A

5 L Stock A
(orange)

+

B

5 L Stock B
(blue)

=

1000
L

fill with water

Then check: **EC** = this week's number · **pH 5.8–6.3** · **Done ✓**

- Grow → EC 2.1
- Flower → 2.4
- Sizing → 2.55
- Ripen → 2.7

EACH MIX AUTO-DELIVERS THESE PPM — YOU DON'T SET THEM

	Grow	Flower	Sizing	Ripen
N	180	175	158	138
K	195	240	285	325
Ca	165	195	195	170

Full N-P-K-Ca-Mg table in Reference. K climbs to build sugar; Ca stays high against BER.

Fruiting & ripening only: also tip in **this week's SOP bag** (green) before topping up.

Always **5 L Stock A + 5 L Stock B** per 1000 L. Drums are pre-made for the stage, so the feed is always right — add the liquids, tip the SOP bag if it's a fruiting stage, top up, set EC + pH. No calculating.

STAGE 1

Nursery — seed to seedling

7–14 days ·
before HST 0

GERMINATE AT
28–30 °C

SEEDLING EC
0.5 → 1.5

READY AT
2–3 true leaves

1 • Soak the seed

- 1 Warm clean water ~28 °C, 7–9 hours.** This "hydropriming" makes sprouting faster and more even. **Don't go past ~12 h** — seeds need oxygen or they drown.
- 2 Optional but recommended (you have H₂O₂):** first dip seed in ~1% **hydrogen peroxide** for **10–15 min**, then rinse, then do the warm-water soak. It sanitises the seed coat (less damping-off) and lifts vigour — both research-backed.
- 3** Only soak fresh seed you will sow the same day.

✓ Also supported

Warm-water soak = quicker, more uniform germination. A dilute KNO₃ soak helps in *cold* weather — not needed in your warm nursery, so plain warm water (± the H₂O₂ dip) is all you need.

2 • Pre-sprout in paper towel

Damp (not soggy) towel, sealed in a box, **warm & dark 28–30 °C**. Radicle "legs" show in 2–4 days. Move each seed the moment the root is 5–10 mm — don't let roots grow into the towel.

3 • Into rockwool

Pre-soak cubes at **pH 5.8, EC 0.5**. Set the sprout **radicle-down**, ~1 cm deep. **Top-feed** to keep moist; dome on; 28–30 °C.

4 • Head breaks → light + bottom-feed

When cotyledons pop up: give **strong light immediately** (or seedlings go leggy) and dome off. **Switch to bottom-feed** (fill the tray, cube wicks up) — keeps the surface dry against damping-off.

5 • Grow on & harden

Bottom-feed, raise EC to ~1.0–1.5, pH 5.8–6.2, 25–28 °C day / 20–22 °C night, plenty of light. Harden off 2–3 days before transplant (a little less water, more light/air).

Watch for this stage

- ! Damping-off** (seedling rots at the base) — from wet, cold, still air → bottom-feed, keep warm, airflow, don't overwater.
- ! Leggy seedlings** — too little light after the head breaks → light them hard.
- ! Fungus gnats** — from soggy media → don't overwater.

! Weak / uneven sprouts — discard them; plant only strong, uniform seedlings.

Move on when: seedlings have 2–3 true leaves, sturdy and uniform → ready to transplant —
buffer the coir (Prep) first.

STAGE 2

Prep — get everything ready

Before planting

JOB

Set up

SUBSTRATE

Coir buffered

STOCK

A & B made

1 • Cocopeat & bags

- 1 **Reused coir:** after the water + H₂O₂ wash, rinse well and **rest ~1 day** so peroxide can't burn roots. Check EC — aim under 1.0.
- 2 **Re-buffer every cycle:** soak/drench with calcium nitrate (~1 kg Calcinit / 1000 L), then drain. This stops the coir stealing calcium and flushes sodium.
- 3 Blend in ~20–30% fresh coir or perlite to keep it airy; discard coir from any diseased crop.
- 4 Fill polybags with drainage holes; stand them on lifters ~1 inch off the floor; one 2 L/hr dripper per bag.

⚠ Never skip the re-buffer

Un-buffered coir + your humidity = blossom-end rot from day one.

How to re-buffer — step by step

You need: 3 × 1000 L tanks of water · 3 kg Calcinit total (1 kg per tank) · a cup for the edge pour.

Day 1

- 1 **Put the polybags back in the greenhouse** — each bag in its correct row and number (A01, A02 ... J43), on lifters ~1 inch off the floor.
- 2 **Put one dripper in each bag** — push it into the top of the cocopeat; check it is not blocked.
- 3 **Mix the tank:** fill to 1000 L, add **1 kg Calcinit**, stir until fully dissolved.
- 4 **Run 1:** run the drippers until the tank is empty. Check water drips out the bottom of the bags — and check the bags at the **far end of every row**. If they aren't wet, tell the supervisor.
- 5 **Wait 2–3 hours** — this is when the calcium sticks to the cocopeat. Don't skip it.
- 6 **Run 2:** mix a new tank (1000 L + 1 kg Calcinit) and run until empty.

Day 2

- 1 **Run 3:** mix a new tank (1000 L + 1 kg Calcinit) and run until empty.

2

Edge pour — the last job: keep ~300 ml of the same **Calcinit solution** for each bag. With a cup, pour it slowly **around the outside edge of the cocopeat**, near the rim — **not** in the middle where the dripper already went.

↑ **Why the edge pour matters**

The dripper only wets a cone under itself — the **edges and corners stay dry**. Roots grow there too. If the edges get no calcium, you get blossom-end rot on the fruit.

⚠ **Never rinse with plain water afterwards**

Plain water washes the calcium back out and ruins the whole job. **Then wait 2 days before transplanting** the seedlings.

2 • Ready the house

Weed-mat dry floor · fans running 24/7 · roof vents open · space bags for airflow · sanitise tools, drippers and structure before a new crop.

3 • Make the feed stock

Make up **Stock A** (orange) and **Stock B** (blue) for the stage — grams in **Reference**. Very strong — just **5 L of each per 1000 L**. For fruiting/ripening stages also pre-weigh the **SOP bags** (tipped straight into the tank).

Watch for this stage

! **Carry-over pests & disease** from the last crop → clean and sanitise everything; discard any bad coir.

Move on when: coir buffered ($EC < 1.0$), bags & drippers set, fans/vents running, house clean → ready to transplant, then Trichoderma drench.

Trichoderma — Cup Drench

By cup, by hand, one bag at a time.

↑ Transplant day

Gently transplant each seedling into its buffered polybag and firm the coir. Then do the Trichoderma drench below. **Plain water only today — no nutrient.** Nutrient feed starts tomorrow morning (HST 0).

⚠ **NEVER put Trichoderma in the drippers or the 1000 L feed tank**

Living spores in a powder — they block the 2 L/hr drippers and the filter, and the plants stop getting water. Do it by cup, by hand, one bag at a time.

1 • Mix it

- 1 Clean 10 L bucket. Fill with **PLAIN water only** — no nutrient.
- 2 Add the Trichoderma. Stir well for 1 minute.
- 3 Stand 15–30 min in the shade, then stir again.
- 4 Use it the **SAME DAY**. Never keep overnight. Never leave the bucket in the sun.

🕒 **Dose: 1 gram per 1 litre → 10 grams in the 10 L bucket**

Weigh it. Do not guess.

2 • Pour it

- 1 Water the bags normally **FIRST**, so the coir is already damp.
- 2 Pour **ONE CUP around the base of the stem**, in a circle. Do not splash the leaves.
- 3 Stir the bucket again every 10 bags, or the powder sinks to the bottom.
- 4 After the drench, do not run a big irrigation for the rest of the day, or it washes out.

Amounts

ONE CUP 250 mL per bag	ONE BUCKET 10 L = 40 bags	WHERE BASE stem base
-------------------------------------	--	-----------------------------------

Never do this

- Never in the drippers or the feed tank.
- Never mix with H₂O₂ (peroxide), chlorine, fungicide, or any chemical — they kill it.
- Never on the same day as the Kalsika silicon drench — leave 2 days between.

When to do it

Time of day	Early morning or late afternoon only. Never in strong sun.
How often	First: 3–5 days after transplant. Then every 2–3 weeks. Write the date on the board.
After H₂O₂	Coir must rest 1 day first.

If unclear, STOP and ask the supervisor.

PRE-WEIGHED BAGS · ONE PER STAGE

One bag per stage

Everything is already in the bag. Nothing to add at mixing time.

↑ The whole job

Tip **BAG A** into **Drum A**. Tip **BAG B** into **Drum B**. Top both to **50 L** and stir until clear. That is all.

GROW Vegetative HST 0–25 2 sets	BAG GROW-A → DRUM A <table><tr><td>Calcinit Yara</td><td>8.70 kg</td></tr><tr><td>Krista K Plus Yara</td><td>3.45 kg</td></tr><tr><td>bag weight</td><td>12.15 kg</td></tr><tr><td></td><td></td></tr></table>	Calcinit Yara	8.70 kg	Krista K Plus Yara	3.45 kg	bag weight	12.15 kg			BAG GROW-B → DRUM B <table><tr><td>Krista MKP Yara</td><td>2.20 kg</td></tr><tr><td>MAG-S Meroke</td><td>5.70 kg</td></tr><tr><td>Vitaflex Meroke</td><td>0.20 kg</td></tr><tr><td>Micro Fe EDDHA 6% Meroke</td><td>0.05 kg</td></tr><tr><td>bag weight</td><td>8.15 kg</td></tr></table>	Krista MKP Yara	2.20 kg	MAG-S Meroke	5.70 kg	Vitaflex Meroke	0.20 kg	Micro Fe EDDHA 6% Meroke	0.05 kg	bag weight	8.15 kg	SOP PER TANK none
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FLOWER Generative HST 25–42 3 sets	BAG FLOWER-A → DRUM A <table><tr><td>Calcinit Yara</td><td>9.45 kg</td></tr><tr><td>Karate Plus Boroni Meroke</td><td>0.80 kg</td></tr><tr><td>Krista K Plus Yara</td><td>1.15 kg</td></tr><tr><td>bag weight</td><td>11.40 kg</td></tr></table>	Calcinit Yara	9.45 kg	Karate Plus Boroni Meroke	0.80 kg	Krista K Plus Yara	1.15 kg	bag weight	11.40 kg	BAG FLOWER-B → DRUM B <table><tr><td>Krista MKP Yara</td><td>2.30 kg</td></tr><tr><td>MAG-S Meroke</td><td>6.05 kg</td></tr><tr><td>Vitaflex Meroke</td><td>0.20 kg</td></tr><tr><td>Micro Fe EDDHA 6% Meroke</td><td>0.05 kg</td></tr><tr><td>bag weight</td><td>8.60 kg</td></tr></table>	Krista MKP Yara	2.30 kg	MAG-S Meroke	6.05 kg	Vitaflex Meroke	0.20 kg	Micro Fe EDDHA 6% Meroke	0.05 kg	bag weight	8.60 kg	SOP PER TANK 310 g
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SIZING Generative HST 42–60 3 sets	BAG SIZING-A → DRUM A <table><tr><td>Calcinit Yara</td><td>9.45 kg</td></tr><tr><td>Karate Plus Boroni Meroke</td><td>0.80 kg</td></tr><tr><td>bag weight</td><td>10.25 kg</td></tr><tr><td></td><td></td></tr></table>	Calcinit Yara	9.45 kg	Karate Plus Boroni Meroke	0.80 kg	bag weight	10.25 kg			BAG SIZING-B → DRUM B <table><tr><td>Krista MKP Yara</td><td>2.10 kg</td></tr><tr><td>MAG-S Meroke</td><td>6.25 kg</td></tr><tr><td>Vitaflex Meroke</td><td>0.20 kg</td></tr><tr><td>Micro Fe EDDHA 6% Meroke</td><td>0.05 kg</td></tr><tr><td>bag weight</td><td>8.60 kg</td></tr></table>	Krista MKP Yara	2.10 kg	MAG-S Meroke	6.25 kg	Vitaflex Meroke	0.20 kg	Micro Fe EDDHA 6% Meroke	0.05 kg	bag weight	8.60 kg	SOP PER TANK 535 g
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RIPEN Generative HST 60–72 1 set	BAG RIPEN-A → DRUM A <table><tr><td>Calcinit Yara</td><td>8.95 kg</td></tr><tr><td>bag weight</td><td>8.95 kg</td></tr><tr><td></td><td></td></tr></table>	Calcinit Yara	8.95 kg	bag weight	8.95 kg			BAG RIPEN-B → DRUM B <table><tr><td>Krista MKP Yara</td><td>1.85 kg</td></tr><tr><td>MAG-S Meroke</td><td>6.25 kg</td></tr><tr><td>Vitaflex Meroke</td><td>0.20 kg</td></tr><tr><td>Micro Fe EDDHA 6% Meroke</td><td>0.05 kg</td></tr><tr><td>bag weight</td><td>8.35 kg</td></tr></table>	Krista MKP Yara	1.85 kg	MAG-S Meroke	6.25 kg	Vitaflex Meroke	0.20 kg	Micro Fe EDDHA 6% Meroke	0.05 kg	bag weight	8.35 kg	SOP PER TANK 645 g		
Calcinit Yara	8.95 kg																				
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⚠ Three rules

1. A bag marked **A** only ever goes in **Drum A**; a bag marked **B** only ever in **Drum B**. Never swap them. 2. One set = one Drum A + one Drum B = **ten 1000 L tanks**, at 5 L + 5 L per tank. 3. **SOP is never in a bag** — weigh it fresh into each 1000 L tank, using the green column above.

One crop = 9 sets, 18 bags. Weigh out: Calcinit 83.05 kg · MAG-S 54.55 · SOP 28.20 (loose) · Krista MKP 19.45 · Krista K Plus 10.35 · Karate Plus Boroni 4.80 · Vitaflex 1.80 · Micro Fe EDDHA 0.45. Double-bag and seal the Calcinit — it pulls water from the air and sets hard.

Making Stock A @ Stock B

50 litre drum · weigh every product.

DRUM A — orange label · 50 L · 2 products only

- | | | |
|---|---|------------------------------|
| 1 | CALCINIT
Yara Calcinit · calcium nitrate | 8.7 kg
8 kg 700 g |
| 2 | K PLUS
Yara Krista K Plus · potassium nitrate | 3.45 kg
3 kg 450 g |

DRUM B — blue label · 50 L · 4 products

- | | | |
|---|---|-----------------------------|
| 1 | MKP
Yara Krista MKP · mono potassium phosphate | 2.2 kg
2 kg 200 g |
| 2 | MAG-S
Meroke MAG-S · magnesium sulphate | 5.7 kg
5 kg 700 g |
| 3 | VITAFLEX
Meroke Vitaflex · pink bag · use kitchen scale | 200 g
0.2 kg |
| 4 | Fe-EDDHA
iron chelate · use kitchen scale | 50 g
0.05 kg |

⚠ Weigh every product. Do not guess.

How to fill a drum

- 1 Fill the drum **HALF FULL** with warm water.
- 2 Weigh product 1. Pour it in. Stir until the water is **CLEAR**.
- 3 Only when clear, weigh the next product. Pour in. Stir until clear.
- 4 Repeat until every product is in the drum.
- 5 Add water up to the **50 LITRE** mark.
- 6 Write **A or B** and **TODAY'S DATE** on the drum.
- 7 Close the lid. Keep out of the sun. Use within 2 weeks.

Never do this

- Never put a Drum A product into Drum B, or the reverse.
- Never mix Drum A and Drum B together — they meet only in the 1000 L tank.
- Never share a scoop or stirrer between A and B.
- Never add the next product while the water is still cloudy.
- Never put SOP or Kalsika into these drums.

Using the drums

5 L from DRUM A + 5 L from DRUM B → 1000 L water. Water first → add A → stir → add B → stir → top up to 1000 L. Then set **EC 2.1**, check **pH 5.8–6.3**. Use a marked 5 L jug. One drum makes 10 tanks of 1000 L.

STAGE 3 · VEGETATIVE

Establish & grow

HST 0 – 25

FEED EC
2.1

WATER
every 3 h

DRENCH
Silicon weekly

Do

- 1 Transplant the seedling into buffered coir; settle the roots gently.
- 2 Train the vine up its string; remove side-shoots (laterals) **below node 8**; keep nodes 9–12 for fruiting later.
- 3 Start the **silicon root drench weekly** from ~HST 8 (Kalsika 0.3–0.5 g/L in plain water, its own pulse — never in the feed).

Feed & water

Feed: Stock A + Stock B → 1000 L · **EC 2.1** · pH 5.8–6.3 (no SOP bag yet). How-to: see the START HERE card.

Water: establish week — every 4 h, 5 min; then every 3 h, 6 min. Steer by run-off (~15–20%); change only at stage change.

Watch for this stage

- ! **Aphids (kutu daun)** — clusters under leaves, curling; they spread mosaic virus. Hit them early.
- ! **Thrips** — silvery flecks, curled young leaves; also a virus carrier.
- ! **Whitefly** — tiny white flies under leaves.
- ! **Spider mites (tungau)** — fine webbing, speckled leaves (worse when hot/dry).
- ! **Caterpillars (ulat)** — chewed leaves.
- ! **Early powdery / downy mildew** — white powder or yellow patches on leaves.

Do: check ~10 plants per row, 2×/week (look under leaves) · **act at first sighting — don't wait for spread** · yellow & blue sticky traps · prune/remove affected parts · rotate a registered insecticide/fungicide per label (follow the label's safety & re-entry time) · net sides help keep pests out. **First mildew spots or mite stippling → treat that day; keep fruit-fly traps up continuously.**

Move on when: first female flowers appear (~HST 25) → Stage 4. **Make the new S2 stock the night before.**

Watering — Day 0 to Day 25

Two timer settings only.

DAY 0–7 — Establish / root settling

EVERY	MINUTES	PER PLANT
4 h	5	0.5 L
8 am · 12 pm · 4 pm		

▼ ON DAY 8, CHANGE THE TIMER ▼

DAY 8–25 — Grow

EVERY	MINUTES	PER PLANT
3 h	6	0.8 L
8 am · 11 am · 2 pm · 5 pm		

Rules

- Set the timer **ONCE**. Leave it until Day 8.
- Do NOT change the timer because it rained — the coir drains itself.
- Never water at night. All pulses in daylight.
- From Day 8: silicon drench (Kalsika) **once a week**, plain water, on its own — never in the feed tank.

STAGE 3 · RUN-OFF

Check run-off 2–3× a week

Target 15–20% drains from the bottom.

If...	Do
Too little under 15%	Add 1 minute to every pulse.
Too much over 20%	Remove 1 pulse from the day.

Important

Only change it if it is wrong for **several days** — not for one day.

Run-off log

Write the date, day (HST), drain %, and your name — 2–3× each week.

Date	HST	EC	pH	Drain %	Name

Day 25: first female flowers appear → move to Stage 4, change the feed and the timer. **Mix the new S2 stock the night before.**

Silicon (Kalsika) — Weekly Root Drench

Kalsika = potassium silicate, 100 g sachets. To the roots, plain water only. Weekly from Day 8, **stop at Day 60**.

↑ What it does

Silicon builds into the cell walls: **stronger stems, thicker tougher leaves** — aphids, thrips, mites and mildew struggle to get in. Also helps with heat and calcium.

Do it step by step — MANUAL switch

Fresh plain water only — no Stock A, no Stock B, no SOP. Use the manual switch for all three runs; do not change the automatic timer.

- 1** Fill the tank with fresh water to **400 L**. Run **3 minutes** — this pushes the old feed out of the lines.
→ **about 300 L should be left in the tank. If not, stop and tell the supervisor.**
- 2** Put **1 sachet (100 g)** of Kalsika straight into the **300 L** in the tank. Do not dissolve it in a bucket first. Do not add more water. Stir it **in the tank** until the water is **CLEAR** — still cloudy, keep stirring, do not switch on.
- 3** Run **8 minutes** — this waters every polybag.
→ **about 30 L left in the tank.**
- 4** Top up with fresh water to **100 L only** — not more. Run **3 minutes** until the tank is empty — this pushes the Kalsika out of the lines.
- 5** Make the normal feed: fill the tank to **1000 L**, add **5 L Stock A + 5 L Stock B**, stir, set the stage **EC**, check **pH 5.8–6.3**. Write the date on the board.

↑ On a silicon day, set the timer to 2 pulses only

The three runs already give the bags their morning water. So on silicon day run **only the afternoon pulses — 2 pm and 5 pm**. Next morning, back to the normal 4 pulses.

If the drippers cannot be used: 3 g per 10 L bucket, **250 ml at the base of each polybag** (40 bags per bucket, half a day for the farm).

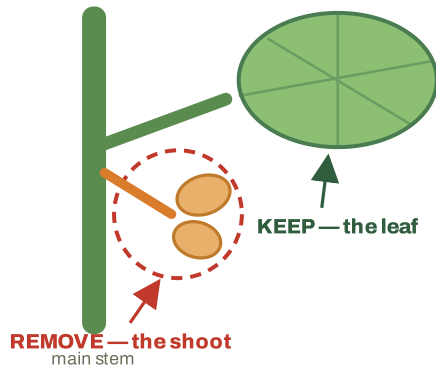
⚠ Never do this

- **Drum A and Drum B — never, for any reason.** In the concentrate drums Kalsika sets into white sludge.
- **The 1000 L tank — one at a time.** Empty of Stock A, B and SOP before Kalsika goes in; empty of Kalsika before the feed goes back. Runs 1 and 3 keep them apart.
- Never the same day as Trichoderma — 2 days between.

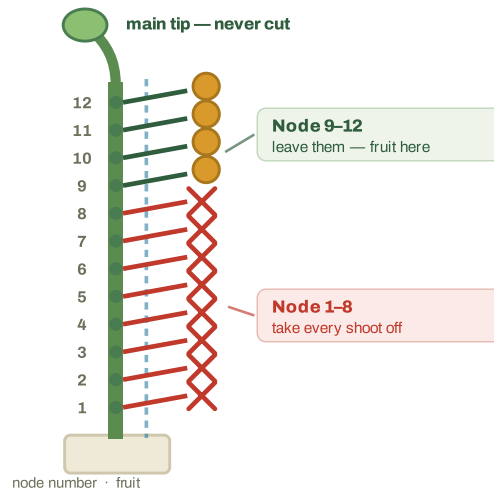
Remove the side shoots

Tunas air — the little shoots in the leaf armpits. Every 2–3 days until node 8.

1 · What to pinch out



2 · How far up the plant



How to do it

- 1 Find the small shoot growing in the **armpit** — where the leaf stalk meets the main stem.
- 2 **Pinch it out with your fingers** while it is small and soft. Do not pull hard, or the stem tears.
- 3 **Take the SHOOT only. Leave the LEAF.** The leaf is feeding the plant.
- 4 Do **not** cut the main growing tip. The leader must keep climbing the string.
- 5 Work in the **morning**, when leaves are dry, so the wound closes before night.
- 6 **No shoots on a plant yet? Leave it alone** — do not go looking or pulling. Check again in **2–3 days**. New shoots keep coming until node 8, around Day 20.

↑ Where to stop

Strip every shoot from node 1 to node 8. From **node 9 upward, leave the shoots alone** — the female flowers, and our fruit, come from those.

⚠ Never do this

Never remove a leaf — a young plant has only 4–6 and needs them all. **Never cut the top tip.** If you cannot tell which is the leader, stop and ask the supervisor. Do not do this on a **leaf-spray** day. The silicon drench goes to the roots, not the leaves — silicon and shoot removal on the same morning is fine.

Carry on the same until Day 25

Nothing changes now. Just repeat the weekly routine below.

↑ The feed does NOT change

S1 Grow — 5 L Stock A + 5 L Stock B per 1000 L. Set **EC 2.1**, check **pH 5.8–6.3**. **No SOP yet** — SOP only starts at Flower. Make a fresh tank whenever it runs out.

↑ The timer does NOT change

Every **3 hours, 6 minutes** — 8 am · 11 am · 2 pm · 5 pm. Leave it alone until Day 25. The only exception is silicon day: **2 pulses only**, 2 pm and 5 pm.

The weekly routine

Job	How often
Silicon drench (Kalsika)	Once a week — see the page before this one
Check run-off is 15–20%	2–3 times a week — not on a silicon day
Trichoderma cup drench	Every 2–3 weeks — 2 days away from silicon
Train the vine, remove laterals below node 8	As the plants grow
Look under the leaves for pests	2 times a week — check ~10 plants per row
Write EC, pH, drain % and your name in the log	Every time you check

⚠ Do not change anything on your own

Do not change the feed, the EC, or the timer because of rain, one dry-looking day, or one odd reading. Only adjust if run-off is wrong for **several days in a row**. If something looks wrong, **tell the supervisor**.

Around Day 25 the first female flowers appear — small melon behind the petals. That is the signal to move to **Stage 4 · Flower**. **Mix the new S2 stock the night before**. Turn the page.

Flower, pollinate & set

FEED EC
2.4

WATER
every 2 h

BORON
via drip (Boroni)

Do

- 1 **Hand-pollinate 8–10 a.m.** on nodes 9–12 — touch the male flower into the female. Do it daily through flowering (wet/humid air makes natural pollination poor).
- 2 Let a few set, then **keep ONE best fruit** per plant (~HST 38–40) and remove the rest.
- 3 **Top the plant** (stop the leader) at ~25 leaves to push energy into the fruit.

Feed & water

Feed: Stock A + Stock B + **SOPbag** → 1000 L · **EC 2.4** (boron rides in on the Boroni — no boron spray).

Water: every 2 h, 7 min. **Spray:** silicon weekly; start **calcium foliar on the young fruit ~HST 35** (Ca-nitrate 4–5 g/L, morning) — preventive because the house stays humid.

Watch for this stage

- ! **Thrips** — worst at flowering; deform flowers & young fruit, carry virus.
- ! **Fruit flies (lalat buah)** — hang methyl-eugenol traps **now**, before fruit forms.
- ! **Poor pollination** — humid, still air → hand-pollinate, keep fans on.
- ! **Gummy Stem Blight** — brown gummy ooze at the stem base in hot+humid air; can wilt/kill the plant. Watch the stem.
- ! **Powdery / downy mildew, aphids.**

Move on when: one fruit is set and starting to swell → Stage 5. **Make the new S3 stock the night before.**

New Stock A & B — Flower

50 litre drum · weigh every product · feed EC 2.4.

↑ Make this the night before Day 25

Mix the new Stock A and Stock B **the evening before**, so the Flower feed is ready first thing in the morning. Do not leave it until the morning of the change.

DRUM A — orange label · 50 L · 3 products

1	CALCINIT Yara Calcinit · calcium nitrate	9.45 kg 9 kg 450 g
2	BORONI Karate Plus Boroni · boron	800 g 0.8 kg
3	K PLUS Yara Krista K Plus · potassium nitrate	1.15 kg 1 kg 150 g

DRUM B — blue label · 50 L · 4 products

1	MKP Yara Krista MKP · mono potassium phosphate	2.3 kg 2 kg 300 g
2	MAG-S Meroke MAG-S · magnesium sulphate	6.05 kg 6 kg 50 g
3	VITAFLEX Meroke Vitaflex · pink bag · use kitchen scale	200 g 0.2 kg
4	Fe-EDDHA iron chelate · use kitchen scale	50 g 0.05 kg

↑ SOP starts now — separate bag, NOT in the drum

Pre-weigh **310 g of SOP per 1000 L** into one bag per tank. Dissolve the bag in a bucket of warm water, then pour it into the 1000 L tank. Never put SOP in Drum A or Drum B.

⚠ Weigh every product. Do not guess.

Same method as the first drum: half-fill with warm water, add **one product at a time** and stir until the water is **CLEAR** before the next one, then top up to the **50 L** mark. Write **A** or **B** and today's date on the drum. Never mix A and B together.

Watering — Day 25 to Day 42

One timer setting for the whole stage.

DAY 25–42 — Flower & fruit set

EVERY	MINUTES	PER PLANT
2 h	7	1.2 L
8 am · 10 am · 12 pm · 2 pm · 4 pm		

Rules

- Set the timer **ONCE** on Day 25. Leave it until Day 42.
- Do NOT change the timer because it rained — the coir drains itself.
- Never water at night. All pulses in daylight.
- Keep the midday pulses — that is when the plant drinks most.
- Silicon drench (Kalsika) **once a week**, plain water, on its own.

↑ Steer by run-off

Check 2–3× a week, aim **15–20%** drain. Under 15% → add **1 minute** to every pulse. Over 20% → remove **1 pulse** from the day. Only change it if it is wrong for **several days**.

Day 42: fruit set and starting to swell → Stage 5, change the feed and the timer. **Make the new S3 stock the night before.**

Hand Pollination

Use the plant's own flowers first. Early morning.

↑ First choice: the plant's own flower

Take the male flower from the **SAME plant** as the female flower. This is the normal way. Do it this way whenever you can.

How to do it

- 1 Work **early morning** (about 6–9 am), when the flowers are freshly open.
- 2 Find an open **female** flower — it has a tiny melon behind the petals. A **male** flower has a thin stem and no melon.
- 3 Take a fresh male flower **from that same plant**. Peel the petals back to show the pollen.
- 4 Brush the pollen gently onto the centre of the female flower. One female = one gentle dab.
- 5 Do **not** use one flower to pollinate many plants. Use a fresh male flower each time.

⚠ If the plant has no male flower open that day

Use the **nearest plant of the SAME variety and the SAME generation** — nothing else.

- **F1 flower → F1 plant only. F3 flower → F3 plant only.**
- Never put F2 or F3 pollen on an F1 plant, or F1 pollen on an F2 or F3 plant.
- Never mix varieties — for example, do **not** use Yellow Kirin Australia F3 (row J) on the Kevin lines.

↑ Write it on the tag

If you used a flower from **another plant** and the fruit sets, write it on the tag using the plant IDs — for example **H18A × H20B**. If it was the plant's own flower, write **self**. This is how we know which seed to keep.

↑ Why it matters

Mixing generations or varieties mixes the genetics of the **seed**. Seed saved from a wrong cross gives uneven plants and fruit next season — different look, size and taste. Keeping it clean keeps the variety **true to type**.

STAGE 5 · GENERATIVE

HST 42 – 60

Fruit sizing

FEED EC
2.55

WATER
every 2 h

FRUIT
1 per plant

Do

- 1 **Support the fruit** in a sling/net tied to the wire so the stalk isn't strained.
- 2 Lightly remove a few old lower leaves for airflow; the fruit's netting develops now.

Feed & water

Feed: Stock A + Stock B + **SOPbag** → 1000 L · **EC 2.55**.

Water: every 2 h, 8 min — peak. **Spray:** calcium foliar weekly to ~HST 55; silicon weekly.

Watch for this stage

- ! **Blossom-end rot (BER)** — sunken brown patch on the fruit base = calcium (your #1 risk). See the picture guide in Reference. Fix: airflow, steady water, don't over-push K.
- ! **Fruit flies** — sting young fruit → maggots & rot inside. Keep traps up; consider **bagging each fruit**.
- ! **Gummy Stem Blight, powdery/downy mildew, anthracnose, fruit rot.**
- ! **Spider mites, aphids.**

Move on when: fruit near full size and netting complete (~HST 60) → Stage 6. **Make the new S4 stock the night before.**

New Stock A & B — Sizing

50 litre drum · weigh every product · feed EC 2.55.

↑ Make this the night before Day 42

Mix the new Stock A and Stock B **the evening before**, so the Sizing feed is ready first thing in the morning. Do not leave it until the morning of the change.

DRUM A — orange label · 50 L · 2 products

- | | | |
|---|--|------------------------------|
| 1 | CALCINIT
Yara Calcinit · calcium nitrate | 9.45 kg
9 kg 450 g |
| 2 | BORONI
Karate Plus Boroni · boron · optional | 800 g
0.8 kg |

DRUM B — blue label · 50 L · 4 products

- | | | |
|---|---|------------------------------|
| 1 | MKP
Yara Krista MKP · mono potassium phosphate | 2.1 kg
2 kg 100 g |
| 2 | MAG-S
Meroke MAG-S · magnesium sulphate | 6.25 kg
6 kg 250 g |
| 3 | VITAFLEX
Meroke Vitaflex · pink bag · use kitchen scale | 200 g
0.2 kg |
| 4 | Fe-EDDHA
iron chelate · use kitchen scale | 50 g
0.05 kg |

Not used this stage: K Plus. Drum A is Calcinit only (plus Boroni if used).

↑ SOP bag — separate, NOT in the drum

Pre-weigh **535 g of SOP per 1000 L** into one bag per tank. Dissolve it in a bucket of warm water, then pour into the 1000 L tank.

⚠ Weigh every product. Do not guess.

Same method as the first drum: half-fill with warm water, add **one product at a time** and stir until the water is **CLEAR** before the next one, then top up to the **50 L** mark. Write **A** or **B** and today's date on the drum. Never mix A and B together.

Watering — Day 42 to Day 60

The wettest stage — six pulses a day.

DAY 42–60 — Fruitsizing

EVERY	MINUTES	PER PLANT
2 h	8	1.6 L
7 am · 9 am · 11 am · 1 pm · 3 pm · 5 pm		

Rules

- Set the timer **ONCE** on Day 42. Leave it until Day 60.
- Do NOT change the timer because it rained — the coir drains itself.
- Never water at night. All pulses in daylight.
- Keep the midday pulses — that is when the plant drinks most.
- This is the heaviest watering of the crop — never let the bag dry out while the fruit is swelling.
- Silicon drench (Kalsika) **once a week** — keep it going through this stage.

↑ Steer by run-off

Check 2–3× a week, aim **15–20%** drain. Under 15% → add **1 minute** to every pulse. Over 20% → remove **1 pulse** from the day. Only change it if it is wrong for **several days**.

Day 60: fruit near full size, netting complete → Stage 6, change the feed and the timer. **Make the new S4 stock the night before.**

STAGE 6 · GENERATIVE

Ripen & brix finish

HST 60 – 80

FEED

S4 → plain water

WATER

taper

TEST

brix ≥ 15

Do

- 1 Feed the ripening mix (low N, high K) to ~HST 68–72, then switch to **plain / low-EC water** for the last 7–10 days to concentrate sugars.
- 2 **Before you switch to plain water, check all four:** fruit near full size · netting fully set · brix rising on the refractometer · no fresh cracking. If any is not ready, hold the ripening feed a few more days.
- 3 Start **brix testing** (~HST 68) with the refractometer.
- 4 Keep water **steady** as you taper — sudden water = cracked fruit. Stop calcium foliar & silicon.

✓ **Before switching to plain water, check**

Fruit at full size & netting set · brix rising on the refractometer · no fresh cracking · run-off EC steady. If it's not there yet, hold the Ripen feed a few more days.

Feed & water

Feed: Stock A + Stock B + **SOPbag** → 1000 L · **EC 2.7**, then plain water for the finish.

Water: every 3 h, 6 min → finish every 4 h, 6 min.

Watch for this stage

- ! **Fruit flies — CRITICAL now.** They attack ripening fruit; keep traps, bag fruit, and destroy any stung fruit (don't leave it in the house).
- ! **Fruit cracking** — from water spikes; keep watering steady, roof keeps rain off.
- ! **Fruit rot / anthracnose, powdery mildew, Gummy Stem Blight** (can kill the plant before ripening → low brix).
- ! **Rodents / birds.**

Move on when: full slip + full net + aroma + **brix ≥ 15** → harvest.

New Stock A & B — Ripen

50 litre drum · weigh every product · feed EC 2.7.

↑ Make this the night before Day 60

Mix the new Stock A and Stock B **the evening before**, so the Ripen feed is ready first thing in the morning. Do not leave it until the morning of the change.

DRUM A — orange label · 50 L · 1 product only

- | | | |
|---|--|------------------------------|
| 1 | CALCINIT
Yara Calcinit · calcium nitrate | 8.95 kg
8 kg 950 g |
|---|--|------------------------------|

DRUM B — blue label · 50 L · 4 products

- | | | |
|---|---|------------------------------|
| 1 | MKP
Yara Krista MKP · mono potassium phosphate | 1.85 kg
1 kg 850 g |
| 2 | MAG-S
Meroke MAG-S · magnesium sulphate | 6.25 kg
6 kg 250 g |
| 3 | VITAFLEX
Meroke Vitaflex · pink bag · use kitchen scale | 200 g
0.2 kg |
| 4 | Fe-EDDHA
iron chelate · use kitchen scale | 50 g
0.05 kg |

Not used this stage: K Plus and Boroni. Drum A is Calcinit only.

↑ SOP bag — separate, NOT in the drum

Pre-weigh **645 g of SOP per 1000 L** into one bag per tank. Dissolve it in a bucket of warm water, then pour into the 1000 L tank.

⚠ Weigh every product. Do not guess.

Same method as the first drum: half-fill with warm water, add **one product at a time** and stir until the water is **CLEAR** before the next one, then top up to the **50 L** mark. Write **A** or **B** and today's date on the drum. Never mix A and B together.

Watering — Day 60 to Harvest

Two settings: dry it down, then plain water.

DAY 60–72 — Ripening · S4 feed · EC 2.7

EVERY	MINUTES	PER PLANT
3 h	6	0.8 L
8 am · 11 am · 2 pm · 5 pm		

▼ ON DAY 72, CHANGE THE TIMER AND STOP THE FEED ▼

DAY 72–HARVEST — Finish · PLAIN WATER, no nutrient

EVERY	MINUTES	PER PLANT
4 h	6	0.6 L
8 am · 12 pm · 4 pm		

Rules

- From Day 72 the tank is **PLAIN WATER only** — no Stock A, no Stock B, no SOP.
- Drying it down is what lifts the sugar (brix). Do not be tempted to water more.
- Never water at night. All pulses in daylight.
- Stop the silicon drench in this stage.

↑ Steer by run-off

Aim **10–15%** drain in this stage — a little drier than before. If leaves wilt badly at midday, add 1 minute; otherwise leave it alone.

Harvest when: full slip + full net + aroma + **brix** ≥ 15.

STAGE 7

Harvest & turnaround

HST 75 – 85+

PASS

brix $\geq$ 15

WEIGHT

1.8–2.2 kg

THEN

reset house

Harvest

- 1 Signs: full netting · ground-spot yellowing · aroma at the stem · half-to-full slip.
- 2 Test one per batch — refractometer on mid-flesh juice (not the centre cavity). **Pass = brix $\geq$ 15, weight 1.8–2.2 kg.** Grade A–D and log it.
- 3 Pick in the cool morning; handle gently; leave a short T-stalk if storing.

Turnaround (back to Stage 2)

Remove the crop & debris → sanitise house, tools, drippers → wash + H₂O₂ the coir → rest → EC check → re-buffer with calcium nitrate. Then you're back at Stage 0.

Watch for

- ! **Carry-over pests/disease** — clean thoroughly; never reuse coir from a diseased crop.

Crop schedule

Three crops a year. Find the week you're in and do that row. Dates follow the SOP— shift $\pm$ a few days to the real transplant date.

Daily always: irrigation + fertigation, EC + pH, pest scouting.

Crop 1 — seed 1 Apr → harvest Jul 2026

Week	HST	Stage	Do this week	EC
Nursery 1–14 Apr	–14→0	Nursery	Seed soak 1 Apr. Warm water 7–9 h → sprout → rockwool → bottom-feed. Harden. 2–3 leaves.	0.5→1.5
1 15–21 Apr	0–6	Establish	Transplant ~15 Apr. Settle roots. Water every 4 h→3 h. Train. Daily EC/pH.	2.1
2 22–28 Apr	7–13	Grow	Train vine. Remove laterals below node 8. Silicon drench weekly. Scout pests.	2.1
3 29 Apr–5 May	14–20	Grow	Keep training. Silicon weekly. Watch aphids/thrips/mildew.	2.1
4 6–12 May	21–27	Grow→Flower	First flowers (~HST 25). Feed EC 2.4 (Boroni in). Hand-pollinate 8–10 AM.	2.1→2.4
5 13–19 May	28–34	Flower/Set	Hand-pollinate daily 8–10 AM. Water every 2 h. Fruit-fly traps up.	2.4
6 20–26 May	35–41	Set/Select	Keep ONE fruit/plant. Top ~25 leaves. Start Ca foliar.	2.4
7 27 May–2 Jun	42–48	Sizing	Feed EC 2.55. Water every 2 h, 8 min. Support fruit. Ca foliar weekly.	2.55
8 3–9 Jun	49–55	Sizing	Netting forms. Ca foliar to ~HST 55. Bag fruit/traps. Watch BER.	2.55
9 10–16 Jun	56–62	Sizing→Ripen	Fruit near full size. Prepare feed switch.	2.55→2.7
10 17–23 Jun	63–69	Ripen	Feed EC 2.7 (cut N, max K). Start brix testing. Steady water.	2.7
11 24–30 Jun	70–76	Ripen/Finish	Plain/low-EC water (last 7–10 d). Stop Ca foliar & silicon. Fruit fly CRITICAL.	water
12 1–8 Jul	77–84	Finish	Plain water. Brix ≥ 15. Watch cracking. Destroy stung fruit.	water
HARVEST 9–16 Jul	85–92	Harvest	Pick fully-ripe only (slip+net+aroma, brix ≥ 15). Sort, pack, sell now.	—

Turnaround (~mid–late Jul): harvest, remove plants, clean & disinfect, flush lines, re-buffer coir, make new stock. Gap to Crop 2 transplant (20 Jul) is tight.

Crop 2 — Jul–Oct 2026

Week	HST	Stage	Do this week	EC
Nursery 6–19 Jul	–14→0	Nursery	Soak 7–9 h → sprout → rockwool → bottom-feed. Harden. 2–3 true leaves.	0.5→1.5
1 20–26 Jul	0–6	Establish	Transplant 20 Jul. Settle roots. Water every 4 h→3 h. Train. Daily EC/pH.	2.1
2 27 Jul–2 Aug	7–13	Grow	Train vine. Remove laterals below node 8. Silicon drench weekly. Scout pests.	2.1
3 3–9 Aug	14–20	Grow	Keep training. Silicon weekly. Watch aphids/thrips/mildew.	2.1
4 10–16 Aug	21–27	Grow→Flower	First flowers (~HST 25). Feed EC 2.4 (Boroni in). Hand-pollinate 8–10 AM.	2.1→2.4
5 17–23 Aug	28–34	Flower/Set	Hand-pollinate daily 8–10 AM. Water every 2 h. Fruit-fly traps up.	2.4
6 24–30 Aug	35–41	Set/Select	Keep ONE fruit/plant. Top ~25 leaves. Start Ca foliar.	2.4
7 31 Aug–6 Sep	42–48	Sizing	Feed EC 2.55. Water every 2 h, 8 min. Support fruit. Ca foliar weekly.	2.55
8 7–13 Sep	49–55	Sizing	Netting forms. Ca foliar to ~HST 55. Bag fruit/traps. Watch BER.	2.55
9 14–20 Sep	56–62	Sizing→Ripen	Fruit near full size. Prepare feed switch.	2.55→2.7
10 21–27 Sep	63–69	Ripen	Feed EC 2.7 (cut N, max K). Start brix testing. Steady water.	2.7
11 28 Sep–4 Oct	70–76	Ripen/Finish	Plain/low-EC water (last 7–10 d). Stop Ca foliar & silicon. Fruit fly CRITICAL.	water
12 5–12 Oct	77–84	Finish	Plain water. Brix ≥ 15. Watch cracking. Destroy stung fruit.	water
HARVEST 13–20 Oct	85–92	Harvest	Pick fully-ripe only (slip+net+aroma, brix ≥ 15). Sort, pack, sell now.	—

Greenhouse prep 21 Oct–7 Nov: remove, disinfect, flush, check equipment, re-buffer, make new stock.

Crop 3 — Idul Fitri (Dec 2026–Mar 2027)

Week	HST	Stage	Do this week	EC
Nursery 20–30 Nov	–14→0	Nursery	Soak 7–9 h → sprout → rockwool → bottom-feed. Harden. 2–3 true leaves.	0.5→1.5
1 1–7 Dec	0–6	Establish	Transplant 1–5 Dec. Settle roots. Water every 4 h→3 h. Train. Daily EC/pH.	2.1
2 8–14 Dec	7–13	Grow	Train vine. Remove laterals below node 8. Silicon drench weekly. Scout pests.	2.1
3 15–21 Dec	14–20	Grow	Keep training. Silicon weekly. Watch aphids/thrips/mildew.	2.1
4 22–28 Dec	21–27	Grow→Flower	First flowers (~HST 25). Feed EC 2.4 (Boroni in). Hand-pollinate 8–10 AM.	2.1→2.4
5 29 Dec–4 Jan	28–34	Flower/Set	Hand-pollinate daily 8–10 AM. Water every 2 h. Fruit-fly traps up.	2.4
6 5–11 Jan	35–41	Set/Select	Keep ONE fruit/plant. Top ~25 leaves. Start Ca foliar.	2.4
7 12–18 Jan	42–48	Sizing	Feed EC 2.55. Water every 2 h, 8 min. Support fruit. Ca foliar weekly.	2.55
8 19–25 Jan	49–55	Sizing	Netting forms. Ca foliar to ~HST 55. Bag fruit/traps. Watch BER.	2.55
9 26 Jan–1 Feb	56–62	Sizing→Ripen	Fruit near full size. Prepare feed switch.	2.55→2.7
10 2–8 Feb	63–69	Ripen	Feed EC 2.7 (cut N, max K). Start brix testing. Steady water.	2.7
11 9–15 Feb	70–76	Ripen/Finish	Plain/low-EC water (last 7–10 d). Stop Ca foliar & silicon. Fruit fly CRITICAL.	water
12 16–23 Feb	77–84	Finish	Plain water. Brix ≥ 15. Watch cracking. Destroy stung fruit.	water
HARVEST 24 Feb–3 Mar	85–92	Harvest	Harvest, sort & pack before 3 Mar — market ≥1 wk before Idul Fitri. Ripe only.	—

 Main rule — no delays

Crop 3 must be harvested, sorted & packed **before 3 March 2027** to reach market ≥ 1 week before Idul Fitri. Report any problem to management immediately.

Reference

Exact numbers for whoever makes the drums & checks the feed. Daily staff don't need this.

GREENHOUSE 1 – MAIN PRODUCTION HOUSE

VARIETY LEGEND

- YELLOW KIRIN KEVIN**
Rows A to E
215 polybags • 430 plants • 430 seeds
- WHITE KIRIN KEVIN**
Rows F to G
86 polybags • 172 plants • 172 seeds
- SPARMANIK MANIS CANDY**
(ORANGE CANDY SELECTION) • Rows H to I
86 polybags • 172 plants • 172 seeds
- YELLOW KIRIN AUSTRALIA F3**
Row J
43 polybags • 86 plants • 86 seeds

GREENHOUSE STATISTICS

Total Rows	10 (A~J)
Polybags per Row	43
Total Polybags	430
Plants per Polybag	2
Total Plants	860
Direction of Numbering	East → West
Row Order	A (South) to J (North)

Each plant is uniquely identified.
Format: **Row + Number + Plant (A/B)**
Example: A01A, A01B, A02A, A02B ... J43A, J43B

IDENTIFICATION SYSTEM

Tag each plant
A = Plant A
B = Plant B

Polybags labeled with number only (e.g. A12)

If one plant dies or has disease:

- Replace only that plant (e.g. A12B)
- The other plant (A12A) keeps its identity and record.

NOTES

- 2 seeds planted per polybag.
- Every plant has an individual ID for recording growth, health and production.
- This system makes selection, pollination and seed collection easier.
- Keep clean and sanitise to prevent infection.

SPACING

Polybag Spacing (Centre to Centre)
625 mm

Row Spacing (Centre to Centre)
1,700 mm

Plant Spacing (within Polybag)
150 – 200 mm

SEED REQUIREMENT

Seeds per Polybag	2
Seeds per Row	86
Total Seeds Required	860

Layout date: 27 May 2025 · Revision: all 10 rows A–J

Greenhouse 1 — Main Production House · 30.0 × 17.0 m growing area · Control room (Ruang Kontrol) & Seedling room (Ruang Seedling) at the east end · numbering South → North (rows), East → West (polybags).

REFERENCE

Greenhouse layout & plant ID

Greenhouse 1 — Main Production House is run as a **production & breeding facility**: every row, polybag and plant has a unique ID so any plant can be tracked seed-to-harvest.

Layout & rules

- **10 rows A–J**, numbered **South → North**; polybags numbered **01–43 East → West** per row.
- **43 polybags per row · 2 plants per polybag → 430 polybags, 860 plants total**.
- **One fruit per plant** — so 2 fruit per bag, each on its own plant.
- **Spacing**: rows 1.70 m (centre-to-centre) · polybags 625 mm · plants 150–200 mm within a bag.

Varieties by row

Rows	Variety	Polybags · seeds
A–E	Yellow Kirin Kevin	215 · 430
F–G	White Kirin Kevin	86 · 172
H–I	Sparmanik Manis Candy (Orange Candy Selection)	86 · 172
J	Yellow Kirin Australia F3	43 · 86

Plant ID system

Label each **polybag with its number only** (e.g. **A12**). Tag the two plants inside as **Plant A** and **Plant B** → full ID **A12A** and **A12B**. Format: *Row + Number + Plant* (A01A, A01B ... J48A, J48B).

If a plant dies or gets sick

Replace **only that plant** (e.g. A12B). The other plant (A12A) **keeps its original ID and record**. Two plants in one bag perform differently — recording each one separately lets you pick the best genetics and cull the poor ones.

Records to keep (per plant)

- **Pollination**: log each hand-cross by plant, e.g. *H18A pollinated by H20B*.
- **Fruit quality**: weight, brix, shape, netting, skin colour, shelf life, disease resistance, yield.
- **Seed selection**: keep seed only from the best-performing plants.
- **Disease**: report by plant ID; check neighbours; record the treatment.

 Staff rules

Always use the correct row + polybag + plant ID. **Never remove labels.** Report disease and record harvest against the right plant ID. Report any missing or damaged label immediately.

REFERENCE

Products

Product	Analysis	Role
Calcinit (Yara)	15.5% N · 19% Ca	Calcium, N — Tank A
Krista K Plus (KNO₃)	13.7% N · 38.4% K	Potassium, N — Tank A
Krista MKP	22.7% P · 28.2% K	Phosphorus — Tank B
Meroke MAG-S	9.6% Mg · 13% S	Magnesium — Tank B
Meroke SOP	~43% K · 18% S	Ripening K (no N) — Tank B
Vitaflex + Fe-EDDHA	Fe·Mn·Zn·Cu·B·Mo	Micros — Tank B
Karate Plus Boroni	Ca-nitrate + 0.3% B	Boron at flowering — Tank A (part-swap)
Kalsika	K-silicate	Silicon — weekly root drench (foliar optional)

REFERENCE

Two tanks • stock • mixing

⚠ Never mix calcium with sulphate/phosphate undiluted

Tank A = Calcinit · Boroni · K Plus. Tank B = MKP · MAG-S · Vitaflex+Fe. They meet only diluted in the 1000 L. **SOP is added separately** (below). Kalsika goes in neither.

Make the stock — large batch in a drum

Recipe key: S1 = Grow (Stage 3) · S2 = Flower (Stage 4) · S3 = Sizing (Stage 5) · S4 = Ripen (Stage 6). Staff pages use the stage & EC; only the mix room uses S1–S4.

5 L of stock feeds 1000 L. So a **50 L drum = ten 1000 L batches**; a **100 L drum = twenty**. Figures below are for a **50 L drum — double every number for a 100 L drum**. Don't make more than ~2 weeks' worth (a giant 1000 L stock drum would sit too long and the iron degrades). Make Tank A and Tank B in *separate* drums.

TANK A · kg / 50 L drum	Grow	Flower	Sizing	Ripen
Calcinit	8.7	9.45	9.45	8.95
Boroni	—	0.8	0.8*	—
K Plus	3.45	1.15	—	—

TANK B · kg / 50 L drum	Grow	Flower	Sizing	Ripen
MKP	2.2	2.3	2.1	1.85
MAG-S	5.7	6.05	6.25	6.25
Vitaflex	0.2	0.2	0.2	0.2
Fe-EDDHA	0.05	0.05	0.05	0.05

To fill a drum: part-fill with warm water, add one product at a time and stir until clear, then top up to the drum mark (50 L or 100 L). Take **5 L to each 1000 L feed tank** with a marked container. *S3 Boroni optional. Keep drums warm; use within 1–2 weeks; keep out of sun. Never combine A & B undiluted.

SOP — separate, add straight into the 1000 L tank

SOP	Grow	Flower	Sizing	Ripen
Per 1000 L (one bag)	—	310 g	535 g	645 g
For a 50 L drum's 10 batches	—	3.1 kg	5.35 kg	6.45 kg

! Why SOP is separate

Potassium sulphate barely dissolves (~110 g/L), so it's the one thing that *can't* go in a strong stock. Kept out of it, Stock A & B stay strong — only **5 L each** per 1000 L. Pre-weigh SOP into per-tank bags; pre-dissolve a bag in a bucket of warm water, then pour into the 1000 L (Flower / Sizing / Ripen only).

✓ Full ingredient list for a drum batch

Tank A: **Calcinit, Boroni, K Plus**. Tank B: **MKP, MAG-S, Vitaflex, Fe-EDDHA**. Separate bag: **SOP**. Separate weekly drench (not in any drum): **Kalsika** (silicon).

Make the feed — per 1000 L

Per 1000 L	Grow	Flower	Sizing	Ripen
Stock A	5 L	5 L	5 L	5 L
Stock B	5 L	5 L	5 L	5 L
SOP bag	—	310 g	535 g	645 g
Set EC	2.0–2.2	2.3–2.5	2.5–2.6	2.6–2.8
Set pH	5.8–6.3	5.8–6.3	5.8–6.3	5.8–6.3

~900 L water → add 5 L A, stir → add 5 L B, stir → (Flower / Sizing / Ripen) tip in the SOP bag, stir → top to 1000 L → set EC then pH.

REFERENCE

PPM targets & ranges

The feed's designed output. Confirm by lab monthly ($\pm 10-15\%$ is fine). Bold = target.

ppm	Grow	Flower	Sizing	Ripen
N	180 165–195	175 160–190	158 145–175	138 120–150
P	50 42–58	52 45–60	48 40–56	42 35–50
K	195 175–215	240 220–260	285 265–305	325 305–345
Ca	165 150–185	195 175–215	195 175–215	170 155–190
Mg	55 48–62	58 50–66	60 52–68	60 52–68

Micros (all stages): Fe 2.2 · Mn 0.30 · Zn 0.33 · B 0.2–0.44 (keep <0.6) · Cu 0.32 · Mo 0.05 ppm.

REFERENCE

Daily checks & log

Check	OK range	If out	✓
Feed EC	stage value	low→add A+B · high→dilute	☐
Feed pH	5.8–6.3	usually ~6.0 by itself; if >6.3 add acid	☐
Drain EC	≤ feed + 1.0	higher→flush more	☐
Drain %	15–20%	low→+1 min · high→–1 pulse	☐

Only change the timer at a stage change (or if run-off drifts for several days). Lab-test the feed monthly.

⚠ Feed lands at pH 6.0 on its own — usually no acid needed

Raw water is pH 7.7, but the fertilisers (especially MKP) pull the mixed feed down to about **pH 6.0** — right in the 5.8–6.3 target. So in normal running you **add no acid**; just confirm with the meter. Keep a little acid on hand only for the odd batch reading **above 6.3** — most likely in ripening (Sizing / Ripen, less MKP) or the dry season. Then add slowly to ~6.0 and note the mL.

💧 Your water (tested)

pH ~7.6 · EC **0.18 mS/cm (184 µS)** · TDS ~86 ppm — soft and low in minerals, so alkalinity is low and the feed self-drops to ~6.0. Because the raw water is so low, nearly all the feed EC is your nutrients — the recipe delivers accurate ppm. **Read the feed in EC/mS mode** and aim for the stage EC. (If reading ppm/0.5 mode instead: ~1050 · 1200 · 1275 · 1350.) Calibrate both pens periodically.

Log

Date	Stage	Feed EC	pH	Drain EC	Drain %	By

REFERENCE

Watering timer (simple intervals)

Set the timer as a plain **every-2/3/4-hour** cycle for the stage, first pulse after sunrise, last well before dark. Small, frequent pulses — keep them running **through midday** (that's when the plant drinks most; don't skip the middle of the day).

Stage (HST)	Every	Min	Clock times	≈/day
Establish (0–7)	4 h	5	8am·12pm·4pm	0.5 L
Grow (8–25)	3 h	6	8am·11am·2pm·5pm	0.8 L
Flower/Set (25–40)	2 h	7	8am·10am·12pm·2pm·4pm	1.2 L
Sizing (40–60)	2 h	8	7am·9am·11am·1pm·3pm·5pm	1.6 L
Ripening (60–72)	3 h	6	8am·11am·2pm·5pm	0.8 L
Finish (72–harvest)	4 h	6	8am·12pm·4pm (plain water)	0.6 L



Set once per stage

Enter the stage's interval & minutes, then leave it. Don't re-program for one rainy day — free-draining coir sheds the extra. Check run-off ~2–3×/week (aim 15–20%); only adjust the whole schedule if it drifts for several days. If the timer has no day/night setting, choose the start so pulses land in daylight — avoid watering into the night.

REFERENCE

Plant problem picture guide

Match what you see. **SEE → MEANS → DO.**



Brown sunken patch on fruit base

Calcium not reaching fruit (BER)

Airflow · steady water · ease early-K



Old leaves yellow between veins

Magnesium low

Check MAG-S · foliar Epsom 10 g/L



New leaves pale, veins green

Iron / pH too high

Lower pH ~6.0 · add Fe-EDDHA



Brown burnt leaf edges

Boron excess / salt high

Check Boroni dose · check EC



White powder on leaves

Powdery mildew

Fans · morning water · fungicide · silicon



Coir always wet, wilting

Overwatering / root rot

Cut a pulse · add perlite · check drainage



Maggots / rot inside fruit

Fruit fly (lalat buah)

Traps · bag fruit · destroy stung fruit



Brown gummy ooze on stem base

Gummy Stem Blight (hot+humid)

Airflow · fungicide · remove infected · avoid stem wetness

REFERENCE

Greenhouse @ climate

Your 30 × 17 m, 7 m tall house — net sides, double netted roof vent, fans 24/7, dry weed-mat floor, raised bags — is a complete passive-vent, low-condensation setup. In this mild, still, cloudy, ~90% climate you **can't reach low humidity**; the air just equals the outside.

✓ **So the goal isn't dry air**

It's: keep leaves dry & moving (fans), keep vents open & net clean, and win on **calcium** (generous Ca, moderate early-K, nitrate-N, steady water, preventive Ca foliar). Optional upgrade: powered gable exhaust fans (cut heat, not humidity below outside).

Quick card

Per 1000 L	Grow	Flower	Sizing	Ripen
Stock A	5 L	5 L	5 L	5 L
Stock B	5 L	5 L	5 L	5 L
SOP bag	—	310	535	645
EC	2.1	2.4	2.55	2.7
Water	3h·6'	2h·7'	2h·8'	3h·6'

Every day: 5 L A + 5 L B (+SOP bag: Flower/Sizing/Ripen) → 1000 L, set EC + pH 5.8–6.3 · check run-off.

Weekly: silicon drench. **Always:** one fruit/plant · fans 24/7 · A & B never mixed undiluted.



Sparmanik Farm

NEW SOP Melon Book · v3.7 · 6 Aug 2026 · Kab. Simalungun, Sumatera Utara

Doses are validated starting points — your EC/pH meters, run-off and brix logs are the final authority. Confirm raw-water chemistry with a lab test each season.